VIVA QUESTION BOOKLET

Comprehensive VSA Study Guide

For Mini Project:

GESTURE CONTROLLED WIRELESS 4WD ROBOTIC CAR

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Contents

Topic 0: Basic Prerequisite Understanding & Knowledge

Q1. What is an embedded system?

Answer: An embedded system is a dedicated computer system designed to perform specific control functions within a larger mechanical or electrical system.

Q2. What does ESP stand for in ESP32?

Answer: ”ESP” stands for Espressif Systems, the semiconductor company that designed and manufactured the chip.

Q3. What do the terms VCC and GND mean on a microcontroller board?

Answer: VCC represents the positive power supply voltage pin, while GND represents the ground or common reference point (0V).

Q4. What does IMU stand for, and what is its main function?

Answer: IMU stands for Inertial Measurement Unit; it is an electronic device that measures a body’s specific force, angular rate, and orientation.

Q5. What does the term ”6-axis” mean in the MPU6050?

Answer: It means the sensor has a 3-axis accelerometer and a 3-axis gyroscope combined on a single chip to track 6 degrees of motion.

Q6. What is the full form of I2C and who developed it?

Answer: I2C stands for Inter-Integrated Circuit, which is a serial communication bus developed by Philips Semiconductors (now NXP).

Q7. What does the ”NOW” signify in the ESP-NOW protocol name?

Answer: It signifies instantaneous (immediate) data transmission without connection or handshaking delays.

Q8. What is the full form of MAC address?

Answer: MAC stands for Media Access Control, representing the physical hardware address of a device.

Q9. What radio frequency range does ESP-NOW operate on?

Answer: It operates on the standard 2.4 GHz Industrial, Scientific, and Medical (ISM) radio band.

Q10. What does ”DC” stand for in DC motor, and how does it turn?

Answer: ”DC” stands for Direct Current; the motor converts electrical energy from a direct current source into rotational mechanical energy.

Q11. What does the ”N” represent in the L298N motor driver part number?

Answer: The ”N” suffix typically indicates the Multiwatt-15 vertical package style of the L298 integrated circuit.

Q12. What is a short circuit, and why is it dangerous?

Answer: A low-resistance connection between supply and ground that causes excessive current draw, leading to battery overheating or component damage.

Q13. Who invented the Mecanum wheel, and when?

Answer: It was invented by Bengt Ilon, a Swedish engineer, in 1972 while working for the company Mecanum AB.

Q14. What are the rollers on a Mecanum wheel made of?

Answer: The passive rollers are typically made of high-traction rubber or plastic materials to ensure firm friction on the ground surface.

Q15. What does ”kinematics” mean in robotics?

Answer: Kinematics is the study of motion (position, velocity, acceleration) of mechanical systems without considering the forces that cause the motion.

Q16. What is the purpose of calibration in sensors?

Answer: Calibration eliminates static offsets and errors by comparing the sensor output against a known reference standard (e.g. flat gravity reference).

Q17. What is an ultrasonic sensor, and how does it measure distance?

Answer: It emits high-frequency sound waves and measures the time taken for the echo to bounce back from an object to calculate distance.

Q18. What does ”noise” mean in electrical signals?

Answer: Noise refers to unwanted random electrical disturbances that degrade or distort the actual signal values.

Q19. State Ohm’s Law and its mathematical expression.

Answer: Ohm’s Law states that current (I) through a conductor is directly proportional to voltage (V) across it, represented by the equation $V = I \times R$.

Q20. State Kirchhoff’s Voltage and Current Laws (KVL/KCL).

Answer: KVL states that the sum of voltages around a closed loop is zero; KCL states that the total current entering a junction equals the current leaving it.

Topic 1: Embedded Systems & ESP32 Architecture

Q21. What is the clock speed of the ESP32 and how does it compare to Arduino Uno?

Answer: The ESP32 operates at a high clock speed of 240 MHz (dual-core), whereas the Arduino Uno operates at a single-core speed of 16 MHz.

Q22. What is the operating voltage of the ESP32 and is it 5V tolerant?

Answer: The ESP32 operates at 3.3V, and its GPIO pins are not 5V tolerant; supplying 5V can permanently damage the chip.

Q23. What is the difference between Core 0 and Core 1 in the ESP32?

Answer: Core 0 is primarily designated for running Wi-Fi, Bluetooth, and network protocols, while Core 1 handles user application code.

Q24. What is the difference between SRAM and Flash memory in the ESP32?

Answer: SRAM (520 KB) is volatile memory used for temporary data runtime, while Flash memory (4 MB) is non-volatile memory storing the firmware code.

Q25. How do you configure a GPIO pin to send signals to the motor driver in code?

Answer: You use the ‘pinMode(pin, OUTPUT)‘ function in the ‘setup()‘ block to configure the designated pin as a digital output.

Q26. What is PWM and why is it used in motor control?

Answer: Pulse Width Modulation (PWM) varies the duty cycle of a square wave to adjust the average voltage delivered, thereby controlling motor speed.

Q27. What is the purpose of the Boot button and EN button on the ESP32 development board?

Answer: The Boot button pulls GPIO 0 low to place the ESP32 in serial bootloader mode, while the EN (Enable) button resets the chip by pulling the chip enable line low.

Q28. Why does the ESP32 need an onboard USB-to-UART bridge chip (like CP2102 or CH340)?

Answer: It translates USB communication from the PC into TTL serial signals (TX/RX) so the ESP32 bootloader can receive code and transmit Serial data.

Q29. What is the ADC resolution of the ESP32 and how does it compare to Arduino Uno?

Answer: The ESP32 features a 12-bit ADC (values 0–4095), whereas the Arduino Uno has a 10-bit ADC (values 0–1023), providing 4 times higher resolution.

Q30. What is the function of the LDO voltage regulator on the ESP32 development board?

Answer: The LDO (Low-Dropout) regulator steps down the external 5V input from the USB port to a stable 3.3V required to power the ESP32 chip.

Q31. What is FreeRTOS and does the ESP32 run it by default?

Answer: FreeRTOS is a real-time operating system that runs by default on the ESP32, allowing multitasking and task scheduling across its dual cores.

Q32. What is a watchdog timer (WDT) in microcontrollers?

Answer: A hardware timer that automatically resets the microcontroller if the software gets stuck in an infinite loop or freezes.

Q33. How does the ESP32 store Wi-Fi credentials or calibration values when powered off?

Answer: It uses Non-Volatile Storage (NVS) library to save key-value pairs directly in a dedicated partition of its Flash memory.

Q34. What is the difference between hardware interrupts and software polling?

Answer: Polling repeatedly queries a pin state in a loop, while interrupts halt the main execution instantly when a hardware pin changes state.

Q35. What is the difference between GPIO pins that have ADC capabilities and those that do not?

Answer: ADC pins read continuous analog voltages and convert them to digital numbers, whereas standard GPIOs only detect discrete high/low logic states.

Q36. What is capacitive touch sensing on the ESP32?

Answer: The ESP32 integrates internal sensors that detect changes in capacitance caused by finger touch on designated GPIO pins.

Q37. What is DAC on the ESP32 and what is its resolution?

Answer: Digital-to-Analog Converter (DAC) outputs analog voltages; the ESP32 has two independent 8-bit DAC channels (GPIO 25 and 26).

Q38. What does the term ”SoC” stand for and is the ESP32 one?

Answer: SoC stands for System on Chip; the ESP32 is a SoC because it integrates the CPU, memory, and wireless transceiver onto a single silicon die.

Q39. What is the standard voltage level for logical HIGH on the ESP32?

Answer: A logical HIGH on the ESP32 is represented by a voltage level of approximately 3.3V (in contrast to 5V on standard Arduino boards).

Q40. What is the purpose of the EN pin capacitor in the ESP32 auto-program circuit?

Answer: It delays the EN line ramp-up during serial bootloading to give the USB-to-UART bridge enough time to assert the GPIO 0 boot signal.

Topic 2: Wearable Glove & IMU Sensor (MPU6050)

Q41. What physical quantities do the accelerometer and gyroscope in the MPU6050 measure?

Answer: The accelerometer measures linear acceleration along the X, Y, Z axes, while the gyroscope measures angular velocity (rate of rotation) around these axes.

Q42. What communication protocol is used between the ESP32 and the MPU6050?

Answer: The I2C (Inter-Integrated Circuit) synchronous serial protocol is used, requiring only two wires: SDA (data) and SCL (clock).

Q43. What are the default SDA and SCL pins on the ESP32?

Answer: The default I2C pins on the ESP32 are GPIO 21 for SDA and GPIO 22 for SCL.

Q44. Why is the magnetometer missing in MPU6050, and what is its consequence?

Answer: The MPU6050 is a 6-axis sensor lacking a magnetometer, which prevents it from referencing the Earth's magnetic north to calculate absolute Yaw.

Q45. What is the mathematical function used to calculate Pitch and Roll from gravity vectors?

Answer: The trigonometric function 'atan2()' is used to compute the angles of Pitch and Roll from the raw acceleration components.

Q46. What is the purpose of the MPU6050's Digital Motion Processor (DMP)?

Answer: The onboard DMP parses raw sensor data and computes complex orientation outputs, offloading sensor fusion calculations from the ESP32.

Q47. What is the full-scale range of the MPU6050 accelerometer and gyroscope, and how is it configured?

Answer: The accelerometer can be set to $\pm 2g, \pm 4g, \pm 8g, \pm 16g$, and the gyroscope to $\pm 250, \pm 500, \pm 1000, \pm 2000$ by writing to the config registers.

Q48. What is sensor drift in gyroscopes, and how does it affect position tracking over time?

Answer: Gyroscope drift is the accumulation of small bias errors over time during integration, causing calculated angles to steadily wander from actual values.

Q49. What are I2C pull-up resistors and why are they required?

Answer: I2C lines are open-drain; pull-up resistors (typically 4.7kΩ) pull the signal lines high to VCC when no device is pulling them low.

Q50. What are the I2C speed modes supported by the MPU6050?

Answer: It supports Standard Mode (100 kbps) and Fast Mode (400 kbps) to enable rapid data transfer.

Q51. What is the AD0 pin on the MPU6050, and how does it affect the device address?

Answer: The AD0 pin sets the LSB of the I2C address; connecting it to GND sets the address to '0x68', and connecting to VCC sets it to '0x69'.

Q52. What is the difference between raw sensor output and calibrated physical units in MPU6050?

Answer: Raw outputs are 16-bit signed integers; they must be divided by sensitivity scale factors (e.g., 16384 LSB/g) to obtain values in g or $^\circ/s$.

Q53. How does mechanical vibration affect accelerometer readings?

Answer: High-frequency vibrations introduce high-frequency noise spikes, which distort tilt calculations unless filtered out by software or low-pass filters.

Q54. What is the function of the built-in temperature sensor in the MPU6050?

Answer: It measures die temperature to allow software compensation for thermal bias drift in the accelerometer and gyroscope silicon.

Q55. What is the significance of the "FIFO" buffer in the MPU6050?

Answer: First-In-First-Out (FIFO) buffer stores sensor data packets temporarily so the ESP32 can read them in bursts, preventing data loss.

Q56. What is the auxiliary I2C bus on the MPU6050 used for?

Answer: It allows the MPU6050 to directly interface with external sensors (like a magnetometer) without passing data through the main ESP32 I2C bus.

Q57. What is quantization error in sensor ADCs?

Answer: The difference between the continuous physical analog value and its discrete digital representation due to finite resolution limits.

Q58. How does the MPU6050 notify the ESP32 that new data is ready to be read?

Answer: It asserts a digital signal on its INT (Interrupt) pin, triggering a hardware interrupt routine on the ESP32.

Q59. What is the default I2C clock frequency used in your ESP32-MPU6050 communication?

Answer: The communication typically runs at 400 kHz (Fast Mode) for rapid, real-time sensor updates.

Q60. What does ”degrees of freedom” (DoF) mean, and how many does the MPU6050 have?

Answer: DoF represents the independent directions of motion tracked; the MPU6050 has 6-DoF (3 translation axes + 3 rotation axes).

Topic 3: ESP-NOW Wireless Protocol

Q61. What is ESP-NOW and what frequency band does it operate on?

Answer: ESP-NOW is a low-latency, connectionless peer-to-peer wireless protocol developed by Espressif Systems operating on the 2.4 GHz RF band.

Q62. Why is ESP-NOW faster than standard Wi-Fi?

Answer: Unlike standard Wi-Fi, ESP-NOW does not require a router connection or IP handshakes, enabling instantaneous packet transmission.

Q63. What is the maximum payload size of a single ESP-NOW packet?

Answer: A single ESP-NOW packet can transmit a maximum data payload of 250 bytes.

Q64. Why is a MAC address required in ESP-NOW?

Answer: In a connectionless network, the transmitter must target the receiver's physical hardware MAC address to route the data packet directly.

Q65. How do you register a function to handle incoming ESP-NOW packets on the receiver?

Answer: You use the API function 'esp_now_register_recv_cb()' to register a custom callback function that triggers upon packet arrival.

Q66. What is the transmission range and typical latency of ESP-NOW?

Answer: It offers a line-of-sight transmission range of up to 100 meters with an ultra-low latency of only 5 to 10 milliseconds.

Q67. Can ESP-NOW communicate between more than two boards simultaneously?

Answer: Yes, ESP-NOW supports one-to-many and many-to-many topologies, allowing a single transmitter to control multiple receiver cars.

Q68. How does ESP-NOW handle encryption and secure data transmission?

Answer: ESP-NOW supports PMK (Primary Master Key) and LMK (Local Master Key) keys to encrypt packets, though encryption is disabled for lowest latency.

Q69. What happens if the receiver ESP32 is powered off when a packet is sent?

Answer: The transmitter's send callback function returns a failure status ('ESP_NOW_SEND_FAIL') because no ACK frame is received.

Q70. Does ESP-NOW require a Wi-Fi network SSID and password to function?

Answer: No, ESP-NOW uses raw 802.11 vendor-specific action frames, bypasses the Wi-Fi connection phase entirely, and requires no credentials.

Q71. Can ESP-NOW operate while the ESP32 is also connected to a normal Wi-Fi router?

Answer: Yes, ESP32 supports co-existence modes, but both connections must operate on the same Wi-Fi channel (typically channel 1 to 11).

Q72. How does the receiver distinguish packets sent by your transmitter from another glove in the same room?

Answer: The receiver checks the sender MAC address parameter in the callback function to verify it matches the authorized transmitter.

Q73. What is the maximum number of paired peers supported by ESP-NOW?

Answer: ESP-NOW supports up to 20 paired peers, which can be encrypted or unencrypted, depending on the network configuration.

Q74. How does the channel setting affect ESP-NOW transmission stability?

Answer: Selecting a less congested Wi-Fi channel (e.g., channel 6 or 11) reduces RF packet collisions and improves transmission reliability.

Q75. Does ESP-NOW require internet access to transmit commands?

Answer: No, ESP-NOW is a local, peer-to-peer radio protocol that operates offline without any internet connection.

Q76. How does the ESP-NOW protocol handle channel interference?

Answer: It allows manual selection of the 802.11 channel (1 to 14) to avoid bands heavily congested by local Wi-Fi networks.

Q77. What is the broadcast address in ESP-NOW, and what is it used for?

Answer: The broadcast MAC address is ‘FF:FF:FF:FF:FF:FF‘, used to send packets to all listening ESP-NOW devices simultaneously.

Q78. What is the typical packet transmission success rate of ESP-NOW within 10 meters?

Answer: The transmission success rate is typically greater than 99% in clean RF environments due to hardware-level ACK frames.

Q79. What is the power consumption of ESP-NOW compared to standard Wi-Fi?

Answer: ESP-NOW consumes significantly less power because it eliminates the overhead of managing TCP/IP handshake and connection maintenance.

Q80. How does the ESP32 transmitter know if the receiver has moved out of range?

Answer: The registered send callback function returns a send status of ‘ESP_NOW_SEND_FAIL‘ when the receiver fails to reply with an ACK.

Topic 4: H-Bridge Drivers & Power Schemes

Q81. What is an H-Bridge and what is its primary function?

Answer: An H-Bridge is an electronic circuit of four switches that reverses the polarity of voltage across a motor to control its direction.

Q82. What is the operating voltage range and maximum current limit of the L298N driver?

Answer: The L298N can drive motor voltages up to 46V and deliver a peak output current of 2A per channel.

Q83. Why do we need a split power supply for the motors and the microcontrollers?

Answer: It prevents high-current motor load spikes and inductive noise from causing voltage drops that reset the microcontroller.

Q84. What is the function of the onboard 5V regulator on the L298N driver board?

Answer: The regulator (LM7805) steps down the battery voltage (e.g. 7.4V) to a stable 5V output to power the microcontroller.

Q85. What is the consequence of failing to connect the ground of the battery, ESP32, and motor driver?

Answer: The control signals sent from the ESP32 will lack a common reference return path, causing erratic motor behaviors or total control failure.

Q86. Why are flyback diodes used in H-Bridge motor driver circuits?

Answer: They provide a safe path for high-voltage inductive kickback spikes (back-EMF) generated by motor coils to protect driver transistors.

Q87. What is the difference between a linear regulator (like the LM7805 on the L298N) and a buck converter?

Answer: A linear regulator steps down voltage by dissipating excess energy as heat, while a buck converter uses switching transistors and an inductor for high efficiency.

Q88. What is the purpose of the 5V jumper on the L298N motor driver?

Answer: Placing the jumper enables the onboard 5V regulator to power internal logic from the motor power input; it must be removed if input exceeds 12V.

Q89. What is a Bootstrap circuit in high-side MOSFET drivers?

Answer: It uses a capacitor to generate a gate drive voltage higher than the supply rail, allowing N-channel MOSFETs to switch on the high side.

Q90. Why does the L298N require a heatsink while driving BO motors?

Answer: High internal BJT saturation voltage drops generate substantial heat, which must be dissipated by a heatsink to prevent thermal shutdown.

Q91. What is the difference between motor ”coasting” and ”braking” in H-Bridge logic?

Answer: Coasting disables all switches allowing the motor to spin to a stop slowly, while braking shorts the motor terminals to lock it instantly.

Q92. What is a BMS (Battery Management System) and why is it used in Li-Ion packs?

Answer: A protection circuit that prevents over-charging, over-discharging, and short circuits in Lithium cells to prevent thermal runaway.

Q93. What is the purpose of decoupling capacitors near the power pins of the ESP32?

Answer: They act as local charge reservoirs to supply current during transient spikes, preventing local voltage sag and system crashes.

Q94. What is back-EMF and how does it affect the motor driver?

Answer: Back-Electromotive Force is voltage generated by the spinning motor acting as a generator; it opposes the supply and can damage drivers if unsuppressed.

Q95. What is the difference between N-channel and P-channel MOSFETs in motor control?

Answer: N-channel MOSFETs are typically used on the low side because of lower resistance ($R_{DS(on)}$), while P-channel MOSFETs are used on the high side.

Q96. What is cross-conduction (shoot-through) in an H-bridge, and why is it catastrophic?

Answer: When both high-side and low-side switches on the same side turn on simultaneously, shorting VCC to ground and destroying the transistors.

Q97. Why are 0.1μF capacitors soldered directly across motor terminals?

Answer: To suppress high-frequency electromagnetic interference (EMI) and sparking caused by the rotating motor brushes.

Q98. What is the maximum voltage drop across L298N BJTs under a 1A load?

Answer: The voltage drop is typically around 1.8V to 2.0V, resulting in significant power loss and heat dissipation.

Q99. What is a flyback diode's reverse recovery time and why does it matter?

Answer: The time it takes a diode to turn off when voltage is reversed; fast recovery is crucial in high-frequency PWM switching to prevent shorting.

Q100. What is the purpose of the current sensing pins on the L298N chip?

Answer: They allow connecting external resistors to measure motor current by monitoring voltage drops, useful for detecting motor stalls.

Topic 5: Mecanum Wheels & Kinematics

Q101. What is a Mecanum wheel and how does it differ from a standard wheel?

Answer: A Mecanum wheel has passive rollers mounted at a 45-degree angle around its rim, allowing it to generate lateral thrust vectors.

Q102. How does a Mecanum-wheeled robot perform lateral left sliding movement?

Answer: The front-left and back-right wheels rotate in reverse, while the front-right and back-left wheels rotate forward.

Q103. How does the robot slide directly sideways to the right?

Answer: The front-left and back-right wheels rotate forward, while the front-right and back-left wheels rotate in reverse.

Q104. What is a holonomic drive system?

Answer: A system that can move in any direction (x, y translation and yaw rotation) simultaneously and independently of its orientation.

Q105. How does the transmitter map the tilt angles to fit inside a single byte?

Answer: It constrains the pitch/roll angles to [-90, 90] and maps them to a [0, 254] range using the Arduino ‘map()‘ function.

Q106. Why is the center value of the mapped scale 127?

Answer: The value 127 represents the neutral flat position (midpoint of 0 to 254), which indicates a stationary state.

Q107. Why does a Mecanum wheel robot experience lateral slippage, and how can it be minimized?

Answer: The passive angled rollers slide easily on slick floors; slippage is reduced by using high-grip rubber rollers and software acceleration ramping.

Q108. What happens to the direction of force vectors if you mount a Mecanum wheel backwards?

Answer: The 45-degree angled rollers will oppose the proper force direction, causing lateral steering inputs to move the robot diagonally or rotate it.

Q109. What is the difference between Mecanum wheels and standard Omni wheels?

Answer: Mecanum wheels have rollers mounted at a 45-degree angle to the wheel axis, whereas Omni wheel rollers are mounted at 90 degrees.

Q110. What wheel configuration (pattern) must be used on the chassis for Mecanum steering?

Answer: The wheels must form an ”X” shape when viewed from above, meaning the roller axes point toward the center of the robot.

Q111. How does the robot rotate on its axis (yaw rotation) using Mecanum wheels?

Answer: The left side wheels rotate in reverse, while the right side wheels rotate forward (or vice versa).

Q112. What are the diagonal force vectors generated during sideways movement?

Answer: Opposite wheels generate forward-diagonal and backward-diagonal vectors, which cancel out longitudinal forces and add up laterally.

Q113. How does speed mapping affect Mecanum motor control?

Answer: The PWM values must be balanced across all four motors to ensure that mismatched motor strengths do not cause the robot to drift off-course.

Q114. What is holonomic vs non-holonomic movement in mobile robotics?

Answer: Holonomic robots can translate in any direction instantly (like Mecanum), whereas non-holonomic robots must steer and drive (like a standard car).

Q115. How does the angle of Mecanum rollers affect the direction of movement?

Answer: The 45-degree angle resolves the rotation vector into equal longitudinal and lateral forces, allowing smooth sideways translation.

Q116. What is the difference between diagonal forward-left and diagonal forward-right steering vectors?

Answer: Diagonal forward-left drives only the front-right and back-left wheels, while diagonal forward-right drives only front-left and back-right.

Q117. What is a ”skid-steer” system, and how does it compare to Mecanum?

Answer: Skid-steer turns by running one side faster than the other, causing tyre slip, whereas Mecanum translates sideways without wheel slipping.

Q118. Why is chassis weight distribution critical for Mecanum-wheeled robots?

Answer: Uneven weight distribution causes some rollers to slip more than others, distorting the intended lateral movement vector.

Q119. How does the robot brake to a stop using Mecanum forces?

Answer: By executing active reverse torque (briefly reversing all motors) to quickly cancel out forward kinetic momentum.

Q120. Can a Mecanum wheel robot drive on loose sand or gravel?

Answer: No, loose materials jam the passive rollers and prevent the lateral vector cancellation required for sideways translation.

Topic 6: Troubleshooting & Enhancements

Q121. Why do we define a software ”dead zone” (e.g. 100 to 150) in sensor readings?

Answer: The dead zone prevents minor hand shaking or sensor noise from triggering accidental motor movements when the hand is neutral.

Q122. What is the major limitation of the L298N driver and how does it affect battery life?

Answer: The L298N has a high internal voltage drop (1.5V to 2V) across its BJTs, wasting battery energy as heat and lowering motor power.

Q123. What modern component would you use to replace the L298N driver?

Answer: A MOSFET-based motor driver like the TB6612FNG, which offers extremely low resistance and high power efficiency.

Q124. What is a complementary filter, and why is it used for IMU data?

Answer: It combines high-pass filtered gyroscope data and low-pass filtered accelerometer data to obtain a stable, drift-free orientation estimate.

Q125. How does a Kalman filter improve on a complementary filter?

Answer: It uses a recursive statistical algorithm to estimate the true state by weighting sensor measurements and system noise dynamically.

Q126. How would you add obstacle avoidance capabilities to this car?

Answer: Mount an ultrasonic or ToF distance sensor on the front and write code to override control inputs if an object is detected within 20 cm.

Q127. What is the difference between a Complementary Filter and a Kalman Filter in processing overhead?

Answer: A Complementary Filter runs simple scalar equations, while a Kalman Filter requires heavy matrix operations, increasing ESP32 CPU load.

Q128. How does the battery C-rating affect the performance of your robotic car under load?

Answer: The C-rating determines the maximum current discharge rate; a low C-rating causes voltage brownouts when all four DC motors start under load.

Q129. What is gyroscope integration drift, and how do we calibrate it out in software?

Answer: Accumulating static offset biases causes drift; we calibrate it by measuring the average output while stationary and subtracting it as an offset.

Q130. What is the role of the Kalman filter covariance matrix?

Answer: It represents the uncertainty of the system state and measurement noise, allowing the filter to dynamically weight predictions.

Q131. How does a buck-boost converter differ from a buck converter?

Answer: A buck converter only steps down voltage, while a buck-boost converter can either step up or step down voltage as needed.

Q132. What is a logic level shifter and when is it required?

Answer: A circuit that shifts digital signals from 3.3V to 5V (or vice versa) to allow safe communication between mismatched voltage devices.

Q133. How does temperature affect the calibration of your MPU6050 sensor?

Answer: Temperature changes cause thermal expansion in the MEMS structures, shifting the accelerometer and gyroscope offsets dynamically.

Q134. What is the benefit of adding an optical encoder to each Mecanum wheel?

Answer: It provides real-time wheel speed feedback, enabling closed-loop PID control to maintain precise steering vectors despite wheel slippage.

Q135. What is the sampling rate of the MPU6050, and how does it affect signal filtering?

Answer: It can sample up to 8 kHz; higher sampling rates allow digital low-pass filters to remove high-frequency physical noises more effectively.

Q136. How do you implement a digital low-pass filter (DLPF) in the MPU6050 registers?

Answer: By configuring the ‘DLPF_CFG‘ bits in the ‘CONFIG‘ register to set a cutoff frequency (e.g. 42 Hz) to filter out engine rumble.

Q137. What is the purpose of a PID controller in motor speed regulation?

Answer: It continuously calculates an error value and applies Proportional, Integral, and Derivative corrections to maintain target RPM under load.

Q138. What is the main advantage of a Time-of-Flight (ToF) distance sensor over an ultrasonic sensor?

Answer: ToF sensors use laser light to measure distance directly, which is faster and unaffected by sound reflections or target material textures.

Q139. Why does sensor fusion improve tilt estimation compared to using only an accelerometer?

Answer: It combines the accelerometer’s static accuracy with the gyroscope’s rapid response to reject dynamic centrifugal acceleration errors.

Q140. What is thermal runaway in Lithium-Ion battery packs?

Answer: An uncontrolled temperature rise caused by internal shorts or overcharging that triggers exothermic reactions, leading to fire or explosion.